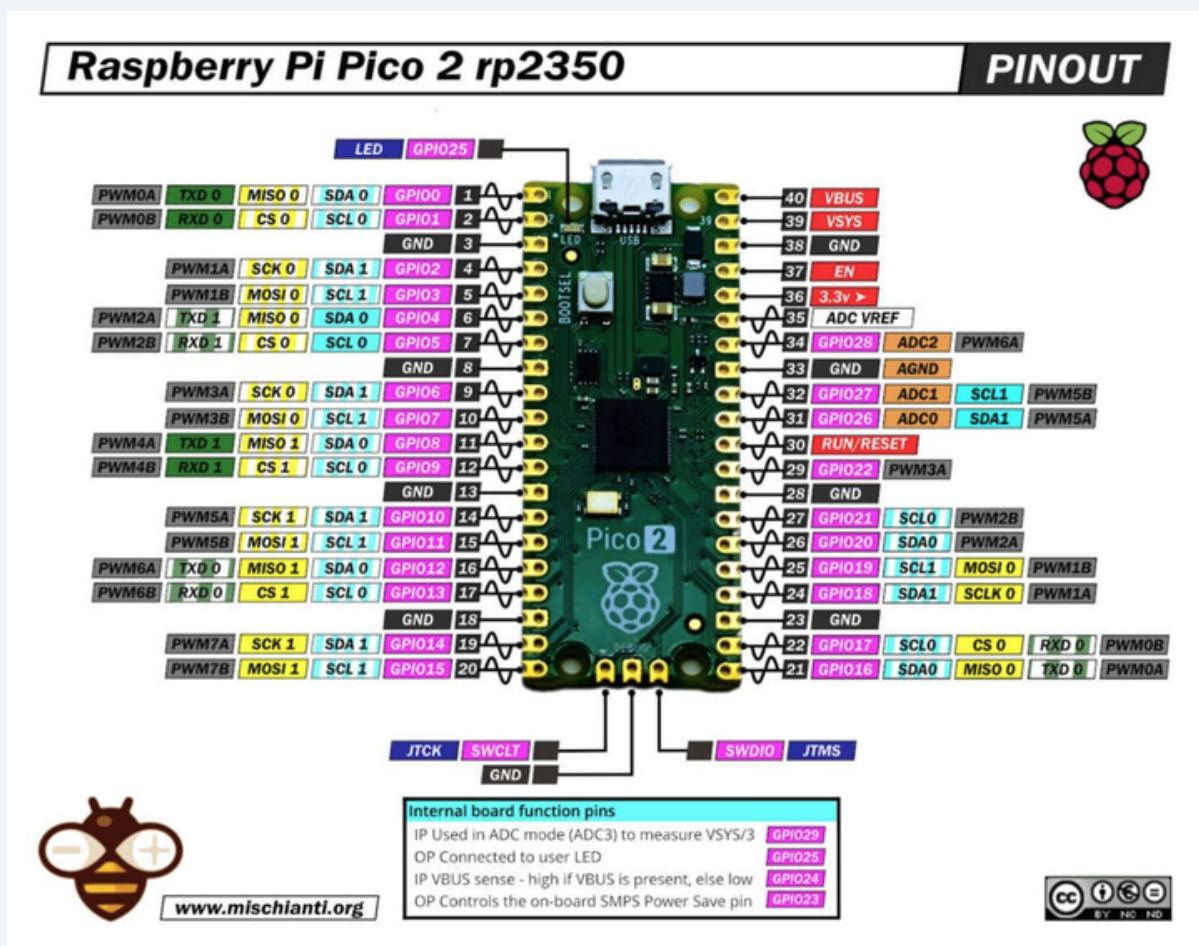
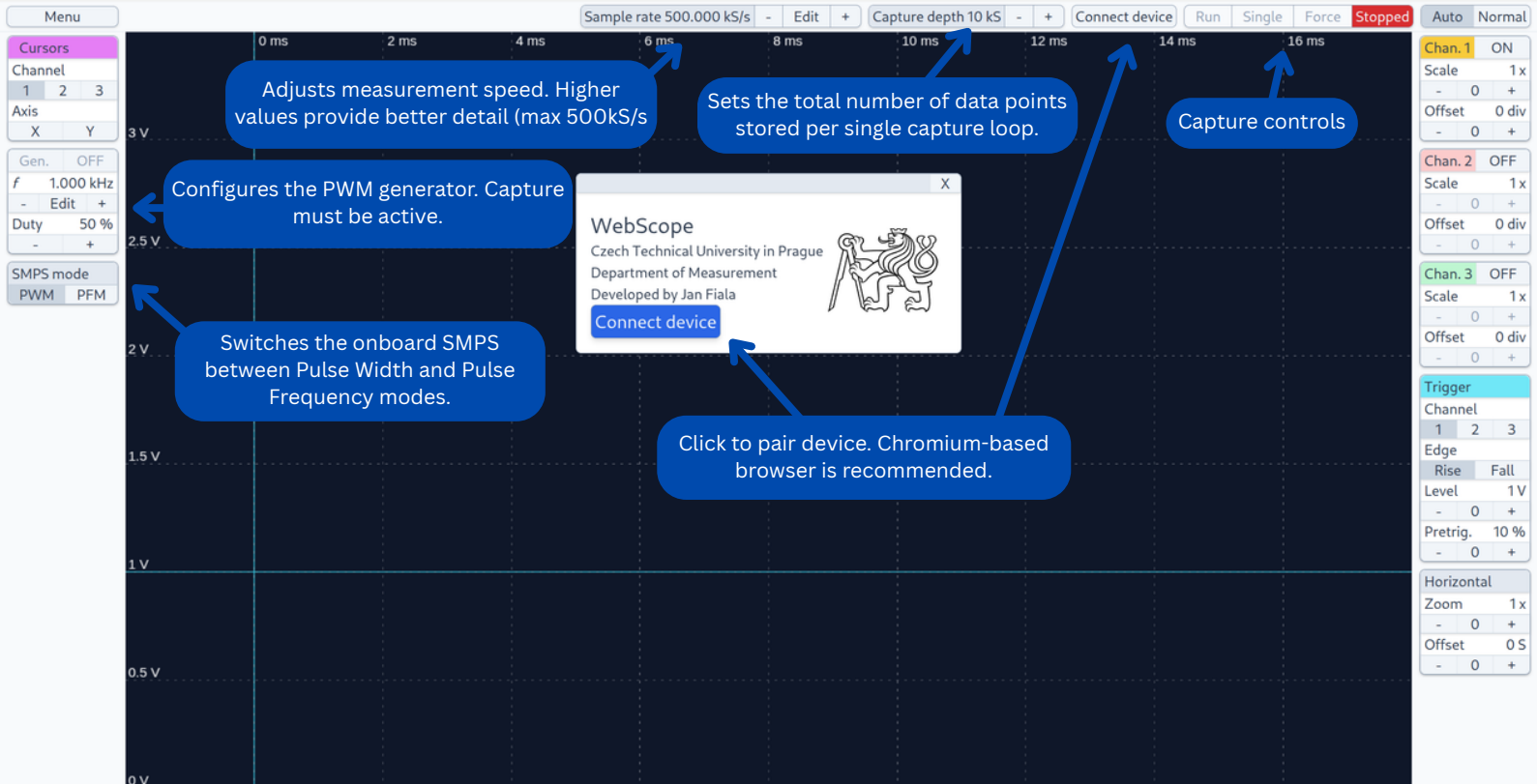




WebScope Hardware Pin Mapping

CH1 (Channel 1 Input): GPIO26 (Physical pin 31, labeled as ADC0)

CH2 (Channel 2 Input): GPIO27 (Physical pin 32, labeled as ADC1)

CH3 (Channel 3 Input): GPIO28 (Physical pin 34, labeled as ADC2)

PWM Output: GPIO16 (Physical pin 21, labeled as PWM0A)

WebUI / Oscilloscope Web Application:

- Accessible online in any Chromium-based browser (MS Edge, Opera, Chrome). Note: Mozilla Firefox is not supported due to a lack of WebUSB API implementation.
- Can also be run locally using the provided offline application package (start by opening the index.html file).
- No need to install third-party desktop drivers or PC applications.

Technical Specifications & Parameters:

- **Sampling Rate:** Up to 500 kS/s for a single channel. When using multiple channels, the sampling capacity is dynamically divided (e.g., 2 channels at 250 kS/s each, etc.).
- **Memory Depth:** Up to 100 kS capture buffer for a single channel (divided dynamically when multiple channels are active, e.g., 2x 50 kS).
- **ADC Resolution:** 12-bit native conversion.
- **Equivalent-Time Sampling (ETS / Stroboscopic):** Up to 1x 48 MS/s or 2x 24 MS/s for repetitive high-frequency signals.
- **Input Voltage Range: 0V to +3.3V. Warning:** Negative voltages or values exceeding +3.3V can cause permanent hardware damage to the MCU pins.
- **Recommended Input Protection:** It is highly recommended to connect a 2k2 (2200 Ohm) resistor in series with each analog input channel to safeguard the pins.

Hardware Pin Configuration & Features:

- **Signal Inputs:**
 - CH1: Physical Pin 31 (GPIO26)
 - CH2: Physical Pin 32 (GPIO27)
 - CH3: Physical Pin 34 (GPIO28)
- **On-board PWM Generator:** Labeled as Pulse Signal Output.
 - Location: Physical Pin 21 (GPIO16)
 - Frequency Range: 8 Hz up to 1 MHz
 - Adjustable Duty Cycle: 0%, 1%, 2%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%